# From LLM Consensus to Cooperative Search

## Weekly Research Progress: LLM Agent Swarms

Chiharu Tomimori

Graduate Research Meeting

October 16, 2026

# Last Week: Research Direction

LLM Proposals $\rightarrow$ Verification

$\rightarrow$ Fault-Tolerant Agreement

Verified Decision

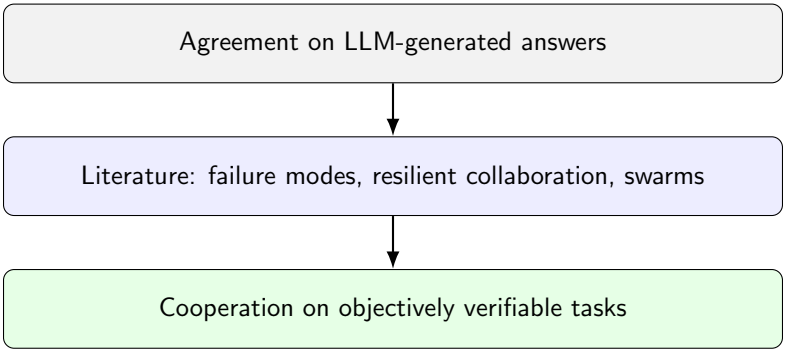
**Unresolved question**

Where is agreement needed, and why?

# Feedback from October 9

- **Motivation:** Why distribute AI agents?
- **Necessity:** What requires agreement?
- **Correctness:** What is a valid outcome?
- **Model:** Which layer, faults, and risks?
- **Position:** What do existing studies solve?

# This Week: Change of Focus




Agreement may be a *mechanism*, not the final objective.

# Background and Motivation

## Why multiple agents?

- Parallel exploration
- Different hypotheses
- Specialized roles

## Why coordinate?

- Reuse discoveries
- Check candidate results
- Avoid wasted work

**Does cooperation outperform independent search?**

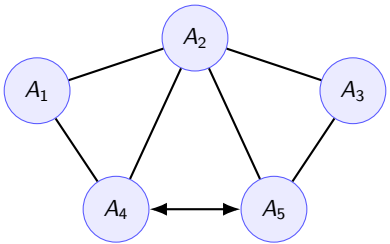
# Related Work: Four Perspectives

| Study | Main focus | What it informs |
|---|---|---|
| Cemri et al. (2025) | Why multi-agent systems fail | Failure taxonomy |
| Huang et al. (2025) | Collaboration with faulty agents | Resilience and topology |
| SwarmBench (2025) | Local swarm coordination | Cooperative tasks |
| Bolzano (2026) | AI-assisted mathematical search | Exploration and verification |

## Current position

Verifiable multi-agent cooperation $\times$ distributed systems

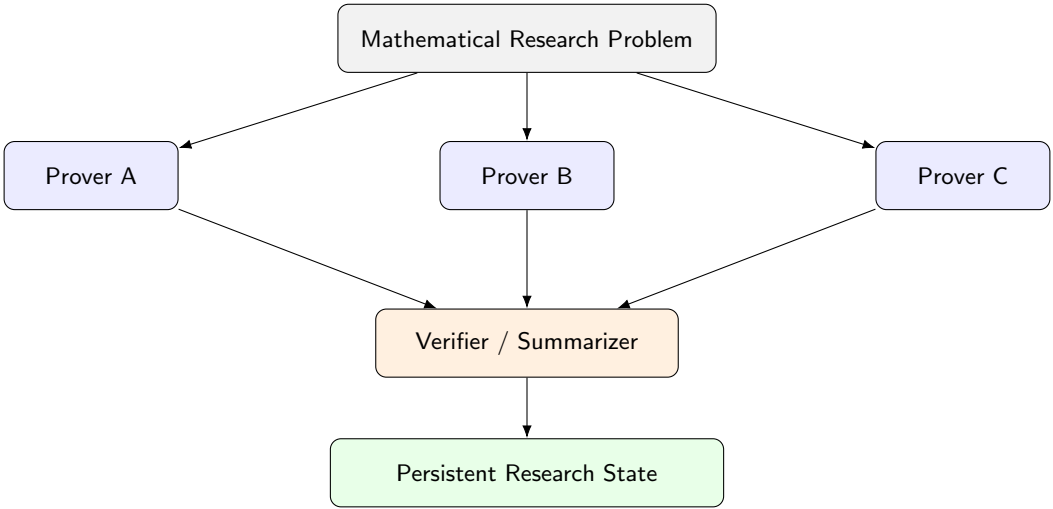
# SwarmBench: Existing Model




Local observations • Neighbor communication • Individual actions

Tasks: Pursuit / Synchronization / Foraging / Flocking / Transport

# Bolzano: Existing Research Architecture




Conceptual architecture; the number of provers varies by setting.

# What Did Bolzano Demonstrate?

| **Supported by the papers** | **Not yet established** |
|---|---|
| AI-assisted proof exploration | Parallelism alone improves accuracy |
| Exploration, verification, and persistent notes | Benefit of state sharing in isolation |
| Reported mathematical research outcomes | A fault-tolerant distributed protocol |

**Interpretation**

Multi-agent architecture $\neq$ proven multi-agent advantage.

# Comparison to Investigate (Not Yet Run)

| Design | Parallel | Roles | Shared state |
|---|---|---|---|
| A: Single agent | No | No | Local only |
| B: Independent agents | Yes | No | No |
| C: Parallel + verifier | Yes | Yes | After rounds |
| D: Cooperative search | Yes | Yes | During search |

## Measurements

Success rate • Time to solution • Token cost • Reuse of discoveries

# Possible Distributed-Systems Problem

**When should agents share intermediate results?**

**Share early**

- Faster reuse
- Risk of unverified claims

**Share after checking**

- Higher confidence
- Extra delay and cost

Hypothesis: verification and propagation timing affect search efficiency.

# Question List and Next Steps

**Open questions**

- When do agents need cooperation?
- What can be verified mechanically?
- What requires a distributed protocol?

**Next week**

- Select a verifiable task
- Compare independent and shared search
- Examine prior distributed methods

**Goal: A concrete research question with a strong baseline**

# References



Cemri et al. *Why Do Multi-Agent LLM Systems Fail?* 2025.
https://arxiv.org/abs/2503.13657


Huang et al. *On the Resilience of LLM-Based Multi-Agent Collaboration with Faulty Agents.* ICML 2025.
https://proceedings.mlr.press/v267/huang25ay.html


*Benchmarking LLMs' Swarm Intelligence* (SwarmBench). 2025.
https://arxiv.org/abs/2505.04364


*Bolzano: Case Studies in LLM-Assisted Mathematical Research.* 2026.
https://arxiv.org/abs/2604.16989


*Bolzano: From Expert-Guided Proof Search to Automated Open-Problem Solving.* 2026.
https://arxiv.org/abs/2610.09769